Assessing Pilot Performance in VR Landing Tasks Using Machine Learning on Physiological Signals

Introduction

- Application of ML remains underexplored in pilot training within the aviation industry [2].
- ML models leveraging physiological signals integrate effectively with legacy training equipment [2].
- ML automates labor-intensive evaluation processes [1].
- Project objective
  - Real-time task performance prediction using machine learning and multimodal physiological data.
  - Supporting the development of adaptive, scalable training systems.

Instrument Landing Tasks

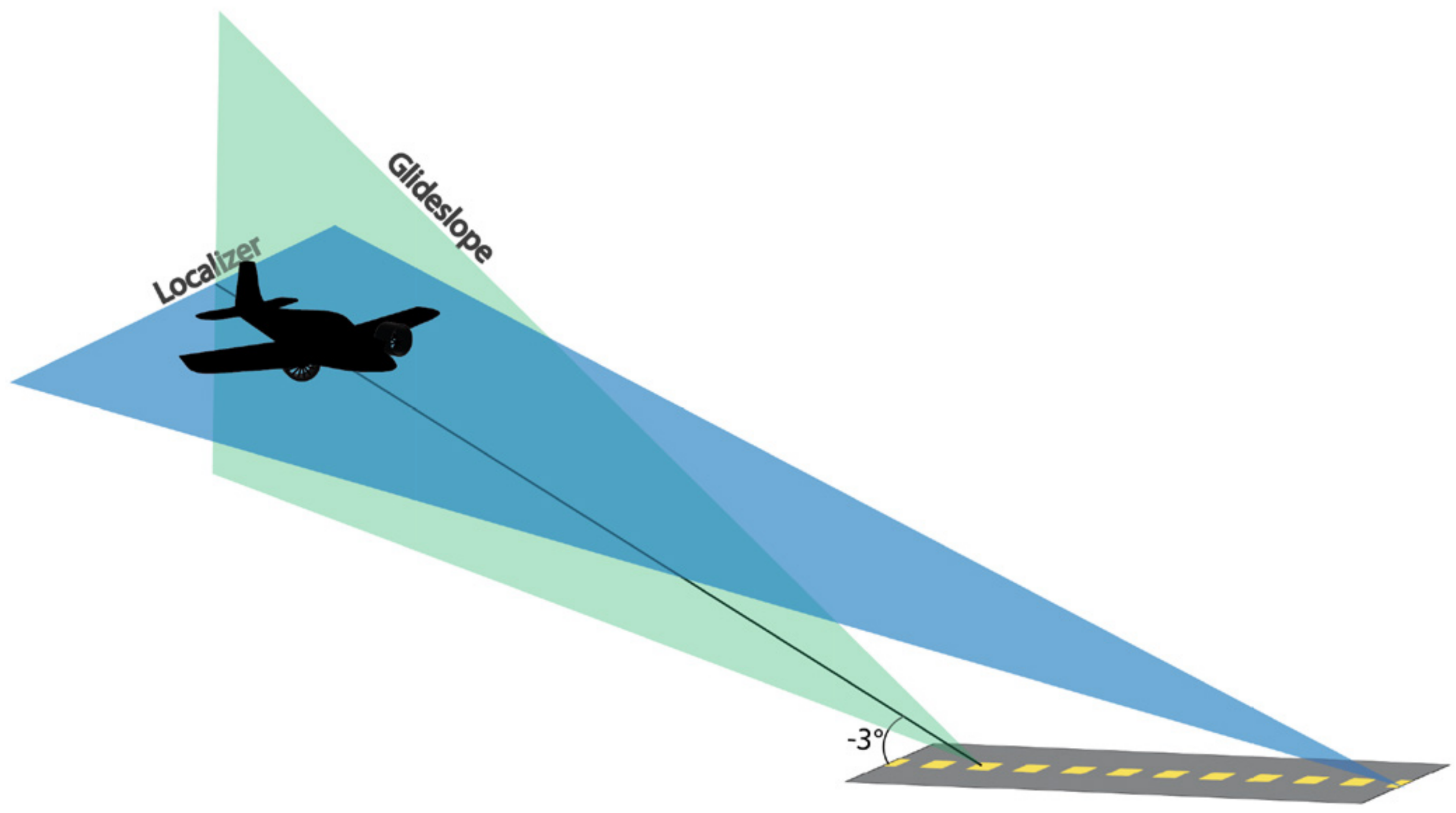


Figure: ILS Approach [3]

Table: Landing Scenarios [7]

Difficulty	Wind	Clouds/Visibility	Turbulence
1	None	No Clouds/Visibility Unlimited	None
2	140° @ 10 kts	Overcast above 2500' Visibility 5 statute miles	None
3	3000': 80° @ 10 kts 1000': 140° @ 10 kts	Overcast above 1000'Visibility 3 statute miles	Light above 1000'
4	4800': 250° @ 10 kts 2800': 80° @ 15 – 20 kts 800': 200° @ 15 kts	Overcast above 400' Visibility 1 statute mile	Light

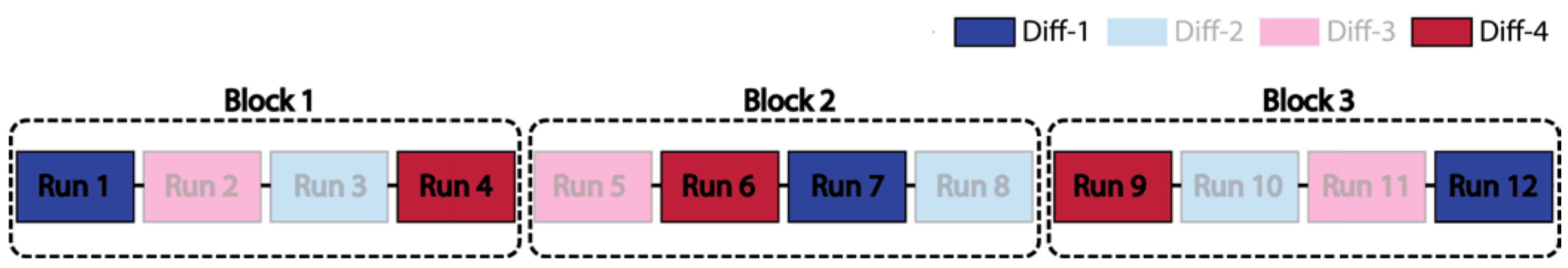
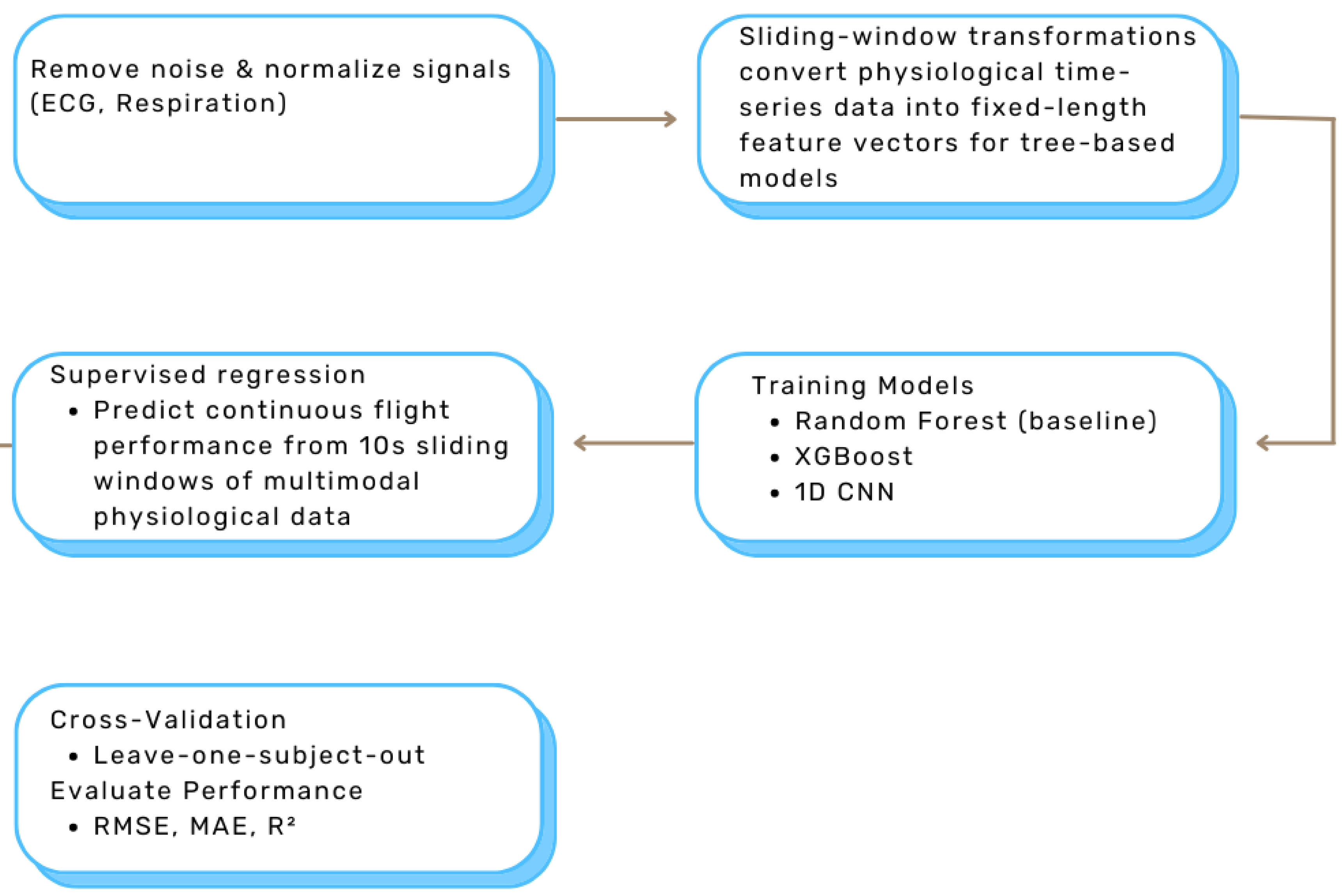


Figure: Order of Difficulty [3]

Methods



Motivation & Background

- Emissions produced during initial training [5]
- Limited training output due to instructor shortage [2]
- VR integration remains relatively underexplored
- Multimodal signals are not widely used in pilot performance analysis [4, 6]
- Difficulty classification tasks achieve an AUC of 0.90 [2, 4]
- Physiological signals correlate with cognitive workload [3]

Dataset

- Dataset: CogPilot [7]

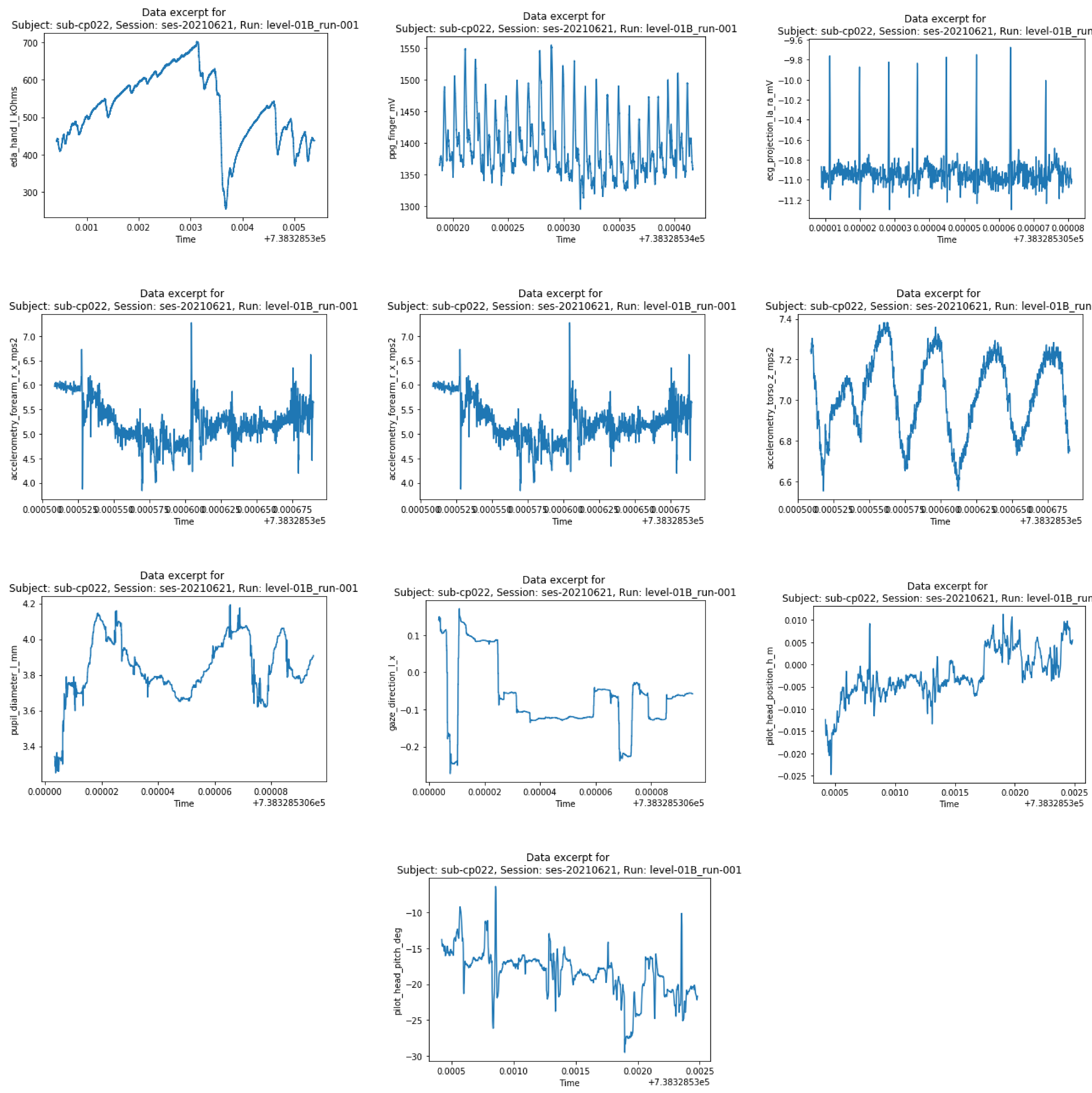


Table: Performance Metrics

Variable Name	Unit	Description
time_dn	datenum	LSL time stamps datenum format
localizer_error	deg	Horizontal angular deviation
glideslope_error	deg	Vertical deviation
ari_speed	knots	Deviation away from 115 knots

References

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